

Wearable Alert Device: User Manual

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Overview

This section describes the details of the CCEW Wearable Alert Device (WAD), provides a block diagram showing the major functional units, and how the WAD's software implementation works with the specific chips on the device.

The WAD's mission is to create a discrete device that could be worn in case of emergency assistance. The overall goal of the WAD is to alert authorities on location, elevation, and an immediate threat such as a gunshot.

The intended audience that we are building the device for is not that of first responders. Instead we are thinking about secondary responders who may not have field training but may still be in dangerous scenarios while working. This device is made for them to have a way of contacting authorities while not giving too much information to the potential assailant.

Features

The following section lists all of the key features of the Wearable Alert Device. For this iteration of the Wearable Alert Device, there are three major components of its functionality. First, the WAD provides wireless data transmission via Bluetooth Low-Energy (BLE) to any Bluetooth Serial messaging mobile app. Second, the WAD contains GPS tracking functionality that records the current coordinates, time, and altitude. Finally, the WAD has the ability to detect gunshots and automatically alert emergency responders when detected.

Wireless Data Transmission

To share with authorities important information of the current situation a secondary responder is in, the WAD wirelessly transmits data using Bluetooth Low-Energy (BLE) protocol. There are two main scenarios that the WAD wirelessly sends data for, where the first scenario requires manual transmission and the second scenario utilizes automatic data transmission. For the first scenario, whenever a secondary responder presses the device's button, the device will transmit data. For the second scenario, if a gunshot is detected by the microphone, the WAD will automatically transmit data. The following table is a list of the data that the WAD sends every transmission.

Table 1: List of data that the Wearable Alert Device transmits.

Type of Data	Unit of Data
Timestamp	“year-month-day hour-minute-second” (CST)
Latitude	Degrees
Longitude	Degrees
Altitude	Meters
Max Audio Reading	dB Sound Pressure Level
Gunshot Detected?	True/False
Call Emergency Responders?	True/False

Whenever these two scenarios occur, the WAD will package all this data together in a little packet and send it off in JavaScript Object Notation (JSON) format. JSON is a standard file format and should be easily readable by any mobile app developed in the future to work with the WAD.

GPS Tracking

The Wearable Alert Device contains GPS tracking technology that constantly records the current time, its current location, and its current altitude. This information is incredibly useful for emergency responders who need to address an urgent situation, allowing them to respond faster.

Gunshot Detection

Containing a regular analog microphone, the WAD can detect gunshots nearby with a fair amount of accuracy. Utilizing some audio signal processing, the WAD has the capability to distinguish gunshots from other loud noises such as loud claps or yelling. As mentioned before, successfully detecting a gunshot will make the WAD automatically transmit data to a mobile device.

User Operation

The following section describes how to properly operate the Wearable Alert Device. Do note that there are certain features, such as the gunshot detection and GPS tracking, that run in the background without the need of manual user operation. Thus, these features will not be discussed. Primarily, the main features that require manual user operation include powering the device, setting up the mobile app, and pressing the button.

Powering the Device

In this section it will be explained how the device is powered on and how to maintain power to the device. Our WAD utilizes two main power supplies to all our chips and modules. The first way is only accessible in our prototype and not in the final device. Simply connect a USB-C power port located on the device to a standard power outlet or source with a USB adapter. This method will then power the rest of the modules, making the device work while plugged into a laptop, for example.



Figure 1: The USB-C Charging Port

The second way of powering our device is by having the battery on the protoboard charged. The device does not have an ‘on - off’ switch meaning that when the battery is connected, the device

will always be running. This means that as long as the battery has charge you will be able to power and use the entire device and its features.

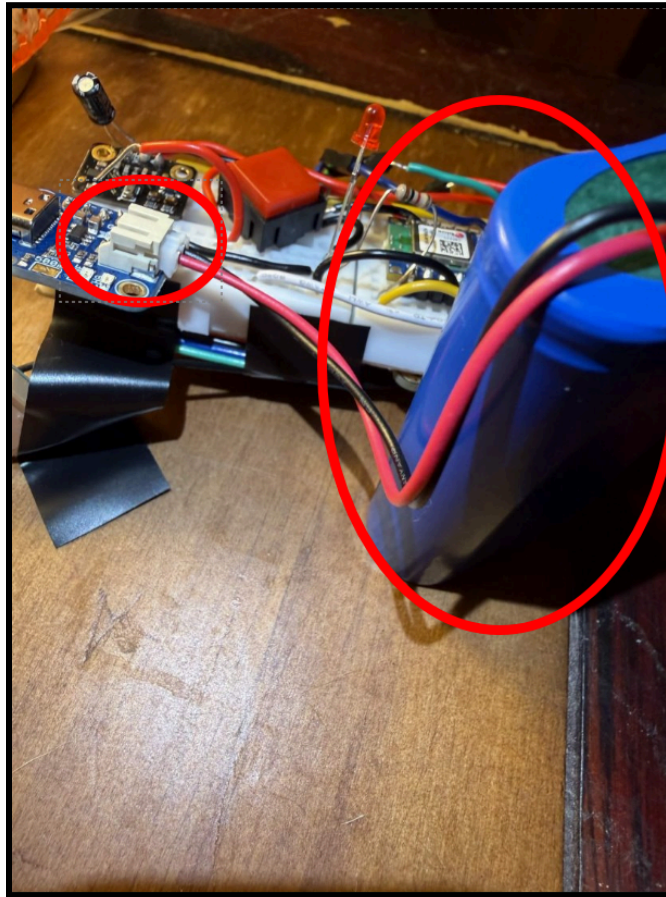


Figure 2: The Battery and the Power Port to Charger

Some key issues to keep in mind when dealing with the powering of the device. Reference the Hardware design document for detailed information on how the battery is charged and where the correct connections are for the prototypes. Since the battery is always plugged in and the device is technically always on, the battery drain is significant and if there are issues even after charging the battery, there may be a dead battery inside the device itself.

Using the nRF Connect for Mobile App

The nRF Connect for Mobile app is a Bluetooth serial messaging app that was used during this project to simulate a mobile app from the CCEW group. Therefore, this user manual is tailored towards usage of this specific Bluetooth serial messaging app. If a different Bluetooth serial messaging app is used, different steps on the mobile app may need to be taken to receive data from a WAD. nRF Connect is available for Android and IOS, and no account is needed, meaning that anyone can download and connect to the WAD. The following steps will take you through setting up this app to receive data from a WAD.

1. Enable Bluetooth on your mobile device.
2. If you do not have it already, download the nRF Connect app onto your mobile device and then open the app.

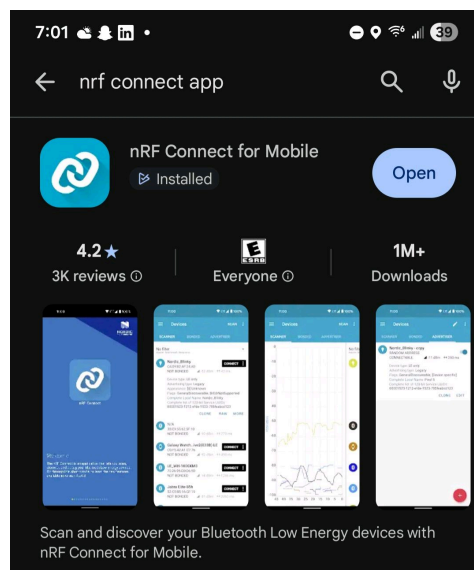


Figure 3: Screenshot of nRF Connect for Mobile app in the Google Play Store on an Android phone.

3. When you first open the app, you will be on the “Devices” tab, where you will be able to see all Bluetooth devices nearby. At the top right, there will be a button that says “Scan”. Select that button to start scanning for Wearable Alert Devices.

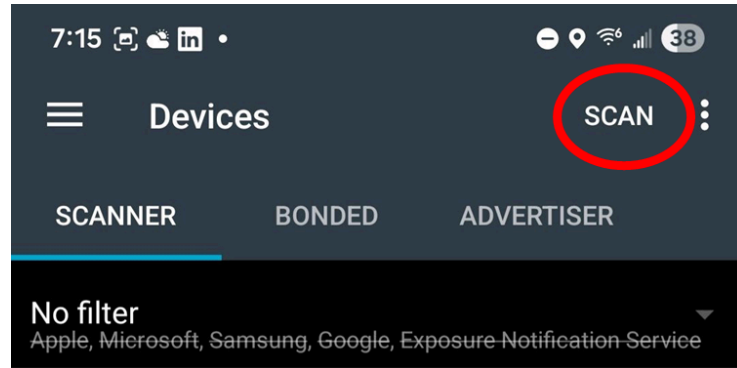


Figure 4: Screenshot of Devices tab in nRF Connect, with the Scan button circled in red.

4. If a powered Wearable Alert Device is nearby, it will appear on the Devices tab as shown below in Figure 5. Select the “Connect” button to connect to the WAD.

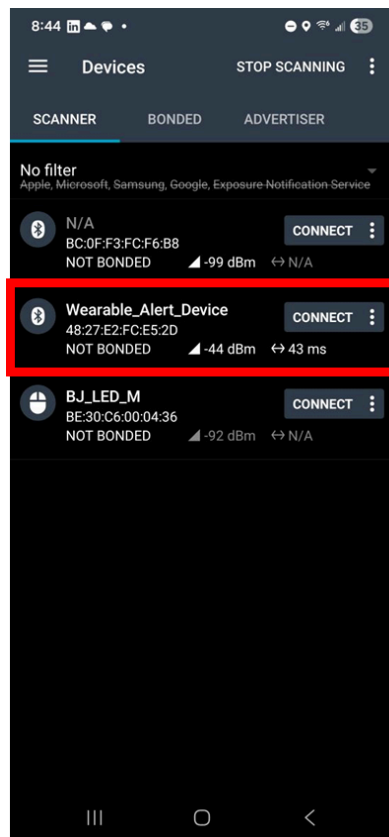


Figure 5: Screenshot of the Wearable Alert Device appearing in the Devices tab.

5. Once connected, you will see this tab for the Wearable Alert Device, as seen in Figure 6.

There are a few settings to configure so that you can actually see the data the WAD transmits.

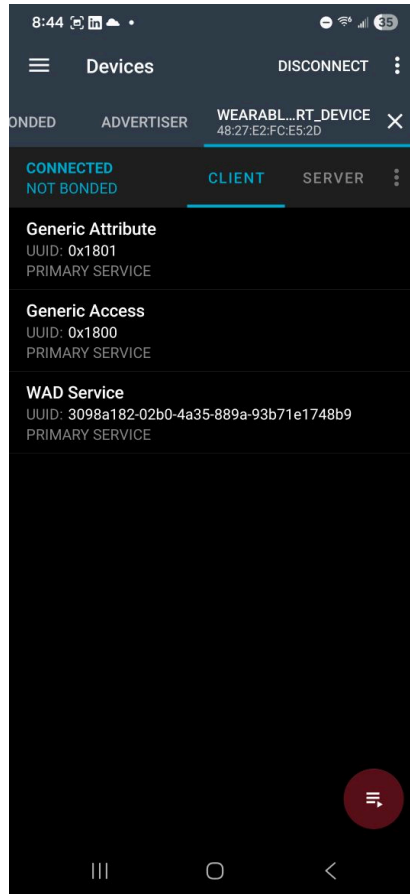


Figure 6: Screenshot of the “Wearable_Alert_Device” tab on nRF Connect app after connecting to a WAD.

6. Select the section labeled “WAD Service” to expand its information (boxed in red in Figure 7). To the right, you will see an icon with three arrows (circled in green in Figure 7). Select this icon to enable live notifications for new WAD data. This icon will be crossed out when it has been selected, and there will be a line of text saying “Notifications enabled” (see Figure 8).

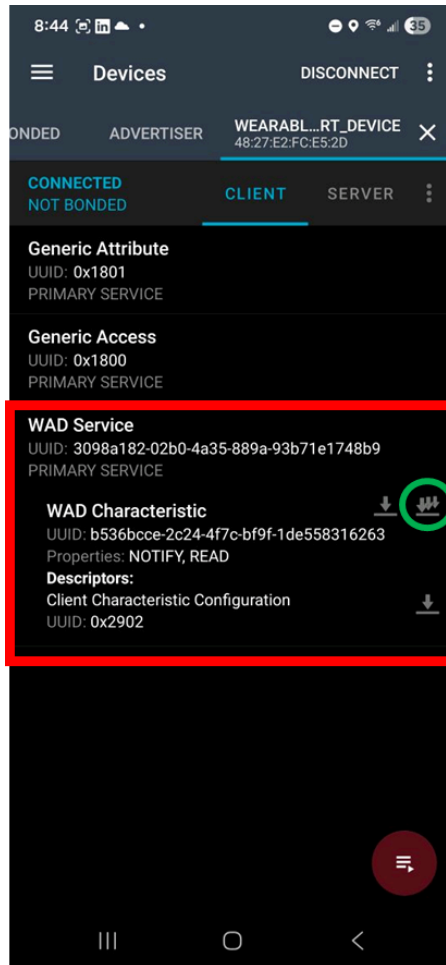


Figure 7: Screenshot of WAD Service section expanded (boxed in red) and the “Enable Notifications” icon (circled in green).

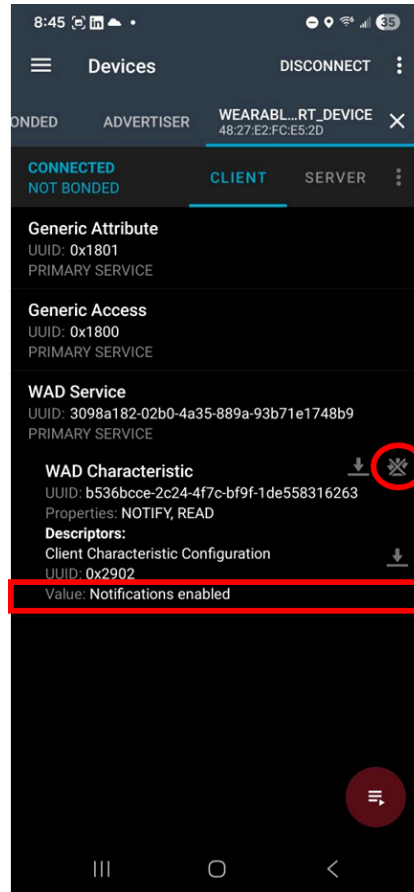


Figure 8: Screenshot of nRF Connect app showing that automatic notifications have been enabled for WAD data (boxed and circled in red).

7. Under that same “WAD Service” section, there is a subsection called “WAD Characteristic.” Press and hold that section until it is highlighted, after which a pencil icon will appear at the top of the app as seen in Figure 9. Select that pencil icon to edit the format of the data when it is received.

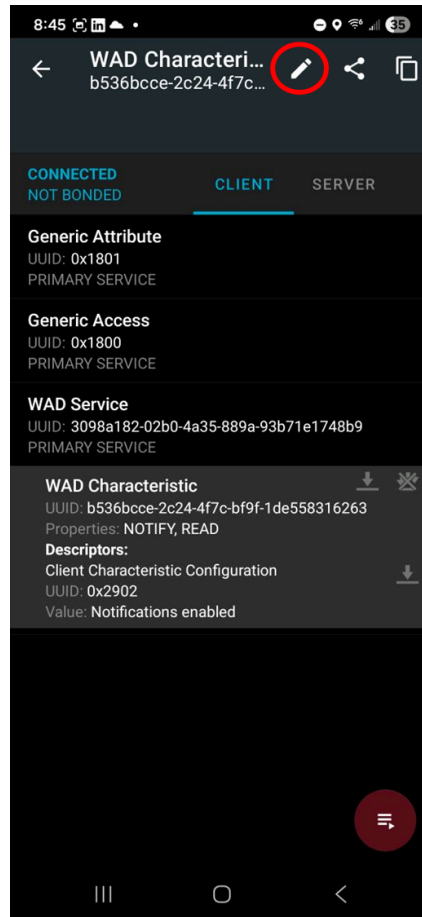


Figure 9: Screenshot of the “WAD Characteristic” being highlighted in gray and the edit option at the top of the app (circled in red).

8. Once the edit option has been selected, you will see a setting called the “Value Format” (see Figure 10). Ensure that this setting is set to the “TEXT” option, not the “NO FORMAT” option. Select “OK” to save this setting.

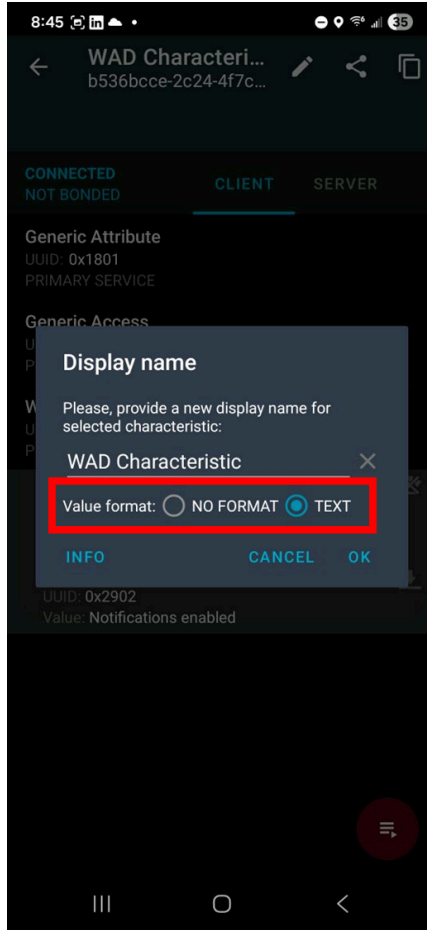


Figure 10: Screenshot of the Value Format setting set to the “TEXT” option (boxed in red).

9. There is one last setting that needs to be adjusted to properly read the data from the WAD. Next to the “Client” and “Server” tabs, there is an icon with three dots that indicates advanced Bluetooth settings (see Figure 11). Select this icon.

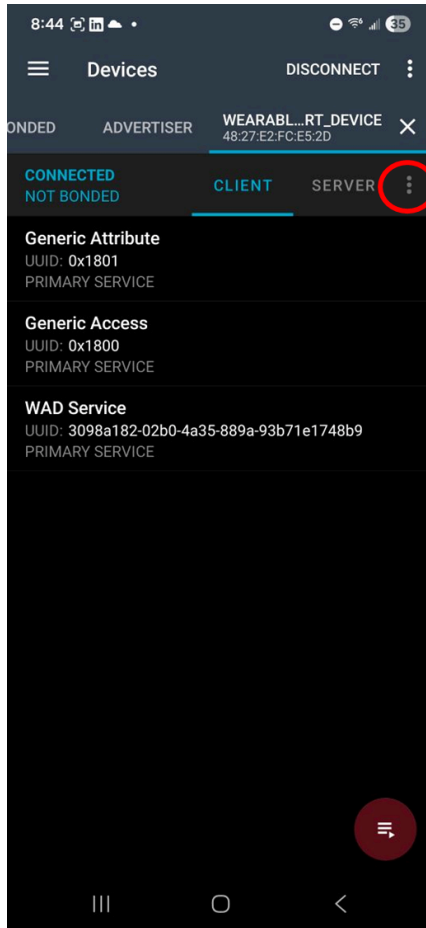


Figure 11: Screenshot of the Wearable Alert Device tab on the nRF Connect app with the advanced Bluetooth settings circled in red.

10. Once the advanced Bluetooth settings have been opened, select the “Request MTU” option (see Figure 12).

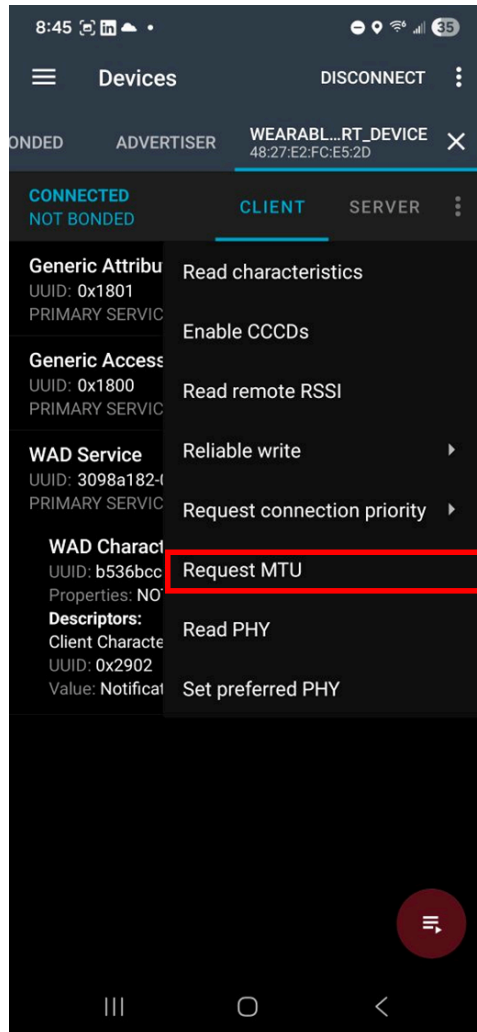


Figure 12: Screenshot of “Request MTU” option in the advanced Bluetooth settings.

11. Selecting the “Request MTU” option will result in a pop-up window that contains a field to change the maximum transfer unit (MTU) (see Figure 13). Ensure this field has the value 517 and select “OK.”

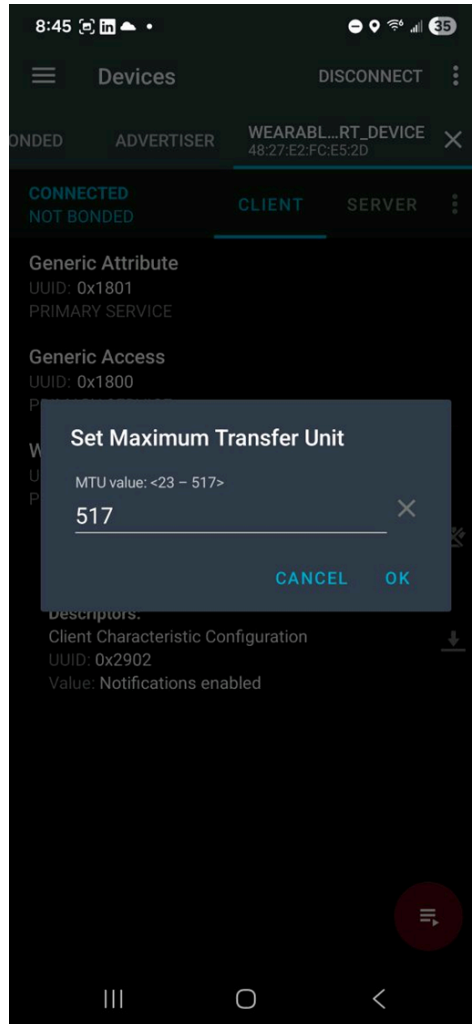


Figure 13: Screenshot of the field to change the MTU, which is set to the value 517.

Finally, the nRF Connect mobile app has been configured to correctly receive data from a WAD. What is important to note is that in an ideal mobile app, these steps would not be needed. These steps are only needed due to the utilization of this specific Bluetooth Serial messaging app. When the WAD transmits data, either due to the press of its button or by detecting a gunshot, the data will automatically appear on the nRF Connect mobile app in its raw JSON format (see Figure 14).

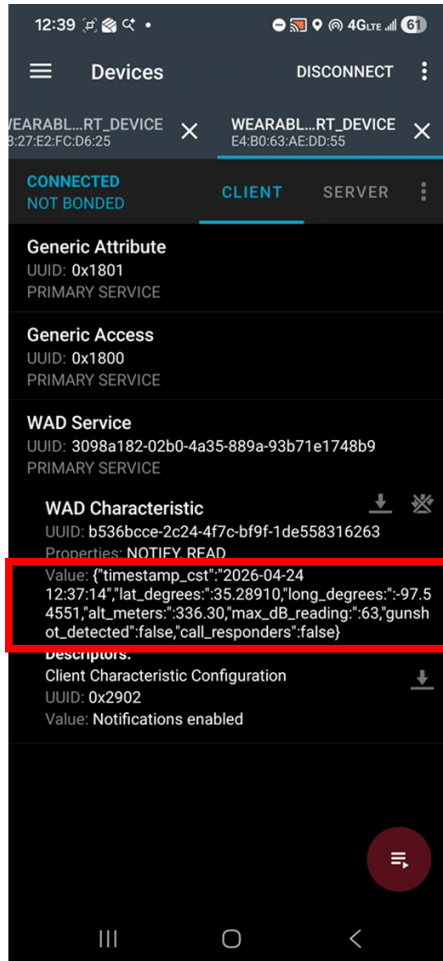


Figure 14: Screenshot of data from the WAD being received by the nRF Connect mobile app.

Button Operation

The button on the top of the device acts as the main communication between the user and authorities.

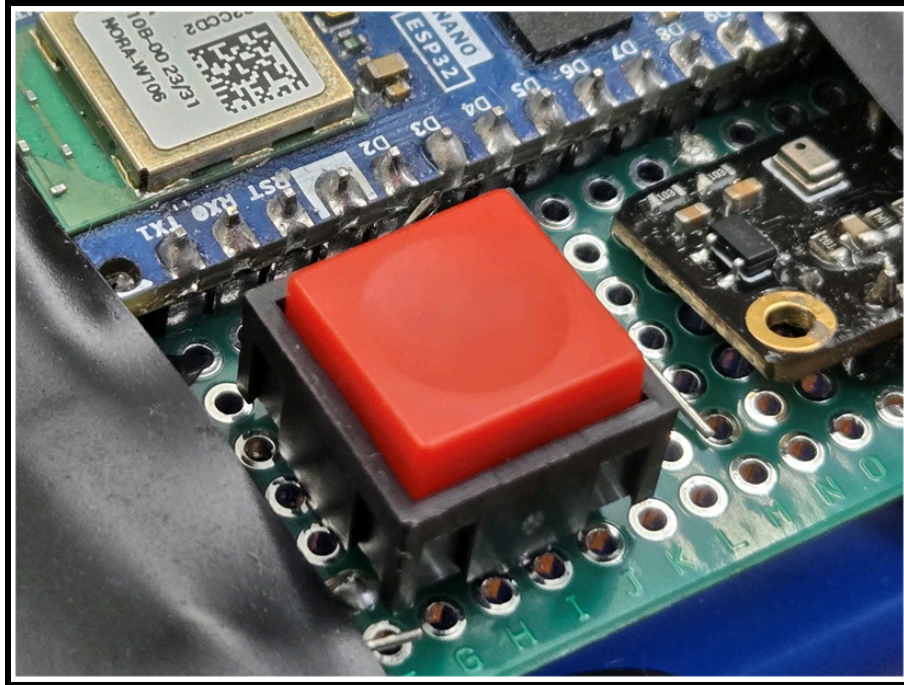


Figure 15: Button on the top of the WAD

As of right now the button will react with a single press of the button and will update the JSON file with the data collected such as GPS and gunshot detection. The button itself will send data to the device if there is a slight holding press. This means that light taps or accidental taps will rarely set the device off. The button can be pressed in rapid succession and the time stamps and location will change with each press. As of right now there is no physical or visual indication that the button has been pressed except for the testing LED on the device, which lights up whenever the button has been pressed. There is future planning for a slight vibration when the button is pressed so that the discrete nature of the device is held intact. However, the only verification methods for a button press are the testing LED and the data being received by the nRF Connect app from the WAD.